

Autonomous Vehicle Simulation (AVS) Laboratory, University of Colorado

Basilisk Technical Memorandum

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POINTING USING A CAMERA-RELATIVE OPNAV DIRECTION AND RATE GYRO INFORMATION

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Scope/Contents

This module provides the attitude guidance output for an opNav pointing mode. Typically this would be to point to a planet center, or a specific heading determined from a camera. The input is an optical navigation direction vector, as well as the body rate information. The output is the standard BSK attitude reference state message. The target direction measurement is crossed with the desired camera axis that is to point at the sun to create a principle rotation vector. The dot product between these two vectors is used to extract the principal rotation angle. With these, a tracking error MRP state is computed. The body rate tracking errors, relative to the inertial frame, are set equal to the measured body rates to bring the vehicle to rest when pointing at the target. Thus, the reference angular rate and acceleration vectors relative to the inertial frame are nominally set to zero. If no heading is available, then the reference rate is set to a camera-fixed value while the attitude tracking error is set to zero.

Rev	Change Description	By	Date
1.0	First Documentation of this module: Nearly Identical to sunSafePoint	T. Teil	2019-09-01
1.1	Principal rotation axis mapped from camera to body frame components; saved heading kept for a first image at 0 s and new images used immediately after a timeout; added non-zero camera mounting, antiparallel heading and saved heading timeout tests	C. Cena	2026-10-27

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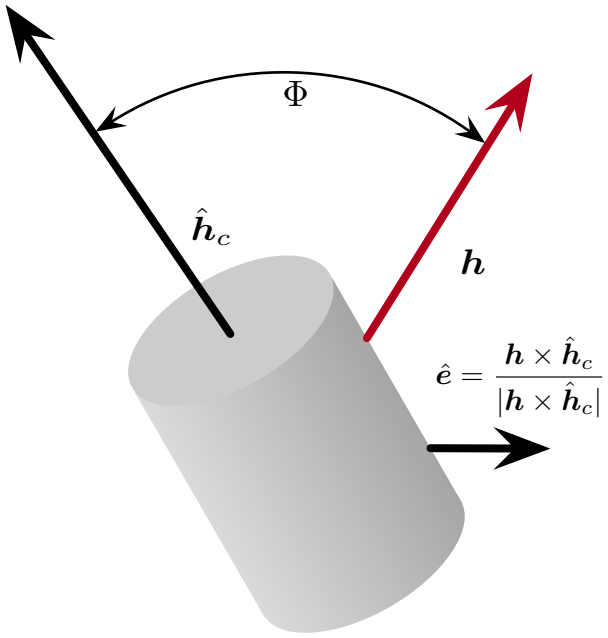


Fig. 1: Body Vector Illustrations.

1 Model Description

1.1 Module Goal

This attitude guidance module has the goal of aligning a commanded camera-fixed spacecraft vector $\hat{\boldsymbol{h}}_c$ with another input vector $\boldsymbol{h}$. If $\hat{\boldsymbol{h}}_c$ is the camera boresight, this module will compute the attitude tracking errors to align the camera towards the target, i.e. achieve relative pointing.

Besides $\boldsymbol{h}$, the second input vector is the inertial body angular velocity vector $\boldsymbol{\omega}_{B/N}$. As the desired planet pointing orientation is unknown, the inertial reference frame acceleration $\dot{\boldsymbol{\omega}}_{R/N}$ is set to zero,.

Note that this module does not establish a unique target-pointing reference frame. Rather, it simply aligns $\hat{\boldsymbol{h}}_c$ with $\boldsymbol{h}$ which is an under-determined 2 degree of freedom condition. Thus, the rotation angle about $\boldsymbol{h}$ is left to be arbitrary in this planet pointing module.

1.2 Equations

1.2.1 Good Planet Direction Vector Case

The planet heading $\boldsymbol{h}$ and the commanded vector $\hat{\boldsymbol{h}}_c$ are both expressed in the camera-fixed frame $\mathcal{C}$. The attitude error of the body relative to the target reference frame $\mathcal{R}$ is written as a principal rotation from $\mathcal{R}$ to $\mathcal{B}$. The axis of this same physical rotation can be expressed in either camera or body frame components without changing its principal angle. The camera-to-reference attitude instead follows the frame composition $[CR] = [CB][BR]$ and generally has a different principal axis and angle. The principal rotation axis of the body attitude error is first computed in camera frame components

$$ {}^{\mathcal{C}}\hat{\boldsymbol{e}} = \frac{{}^{\mathcal{C}}\boldsymbol{h} \times {}^{\mathcal{C}}\hat{\boldsymbol{h}}_c}{|{}^{\mathcal{C}}\boldsymbol{h} \times {}^{\mathcal{C}}\hat{\boldsymbol{h}}_c|} \tag{1} $$

and is then mapped into body frame components using the camera orientation $[CB]$ from the camera configuration message

$$ {}^{\mathcal{B}}\hat{\boldsymbol{e}} = [CB]^T {}^{\mathcal{C}}\hat{\boldsymbol{e}} \tag{2} $$

Note that the planet direction vector $\boldsymbol{h}$ does not have to be a normalized input vector.

The principal rotation angle between the two vectors is given through

$$\Phi = \arccos\left(\frac{\boldsymbol{h}\cdot\hat{\boldsymbol{h}}_c}{|\boldsymbol{h}|}\right) \tag{3}$$

Next, this rotation from $\mathcal{R}$ to $\mathcal{B}$ is written as a set of MRPs through

$$\boldsymbol{\sigma}_{B/R} = \tan\left(\frac{\Phi}{4}\right)^{\mathcal{B}}\hat{\boldsymbol{e}} \tag{4}$$

The set $\boldsymbol{\sigma}_{B/R}$ is the attitude error of the output attitude guidance message. The module variable `opNavMnvrVec` stores ${}^{\mathcal{B}}\hat{\boldsymbol{e}}$.

The rates are also tracked in the body frame. The module allows for a nominal spin rate about the commanded camera axis $\hat{\boldsymbol{h}}_c$ by specifying the module parameter `opNavAxisSpinRate`, in rad/s, called $\dot{\theta}$ in this description. The commanded axis is mapped into body frame components with $[BC] = [CB]^T$, and the nominal spin rate is thus given by

$${}^{\mathcal{B}}\boldsymbol{\omega}_{R/N} = [CB]^T{}^{\mathcal{C}}\hat{\boldsymbol{h}}_c\dot{\theta} \tag{5}$$

Note that this constant nominal spin is only for the case where the planet is visible and the planet-heading vector measurement is available.

If the spacecraft is to be brought to rest, $\boldsymbol{\omega}_{R/N} = \boldsymbol{0}$, then $\dot{\theta}$ should be set to zero. The tracking error angular velocity vector is computed using:

$$\boldsymbol{\omega}_{B/R} = \boldsymbol{\omega}_{B/N} - \boldsymbol{\omega}_{R/N} \tag{6}$$

Finally, the attitude guidance message must specify the inertial reference frame acceleration vector. This is set to zero as the roll about the commanded camera axis $\hat{\boldsymbol{h}}_c$ is assumed to have a constant magnitude and inertial heading.

$$\dot{\boldsymbol{\omega}}_{R/N} = \boldsymbol{0} \tag{7}$$

1.2.2 Saved Planet Direction Vector

Each valid image updates the saved inertial heading ${}^{\mathcal{N}}\boldsymbol{h} = [CN]^{TC}\boldsymbol{h}$ and the time of the last valid image. If the current image is not valid, the saved heading is mapped back into the camera frame with the current attitude, ${}^{\mathcal{C}}\boldsymbol{h} = [CB][BN]^{\mathcal{N}}\boldsymbol{h}$, and used in the equations above. The saved heading is discarded once `timeOut` seconds have passed without a valid image. A valid image is used immediately regardless of how long the previous image is past.

1.2.3 No planet Direction Vector Case

No good planet heading vector is available if the current image is not valid and no saved heading exists, either because no valid image has been received yet or because `timeOut` seconds have passed since the last valid image. A valid image with an all-zero `r_BN_C` vector is also treated as having no heading, as the saved inertial heading is then zero. In these cases the attitude tracking error $\boldsymbol{\sigma}_{B/R}$ is set to zero. Note that the module parameter `minUnitMag` is currently not used by the module, and no norm threshold is applied to $\boldsymbol{h}$.

Further, if the planet is not visible, the module allows for a non-zero body rate to be prescribed. This allows the spacecraft to engage in a constant rate tumble specified through the module configuration vector `omega_RN_B`. In this case, the tracking error rate is evaluated through

$$\boldsymbol{\omega}_{B/R} = \boldsymbol{\omega}_{B/N} - \boldsymbol{\omega}_{R/N} \tag{8}$$

and the output message reference rate is set equal to the prescribed $\boldsymbol{\omega}_{R/N}$ while the reference acceleration vector $\dot{\boldsymbol{\omega}}_{R/N}$ is set to zero.

1.2.4 Collinear Commanded and Planet Heading Vectors

First consider the case where $\boldsymbol{h} \approx \hat{\boldsymbol{h}}_c$. In this case, the cross product in Eq. (1) is not well defined. Let ϵ be a pre-determined small angle. Then, if $\Phi < \epsilon$, the attitude tracking error is set to $\boldsymbol{\sigma}_{B/R} = \mathbf{0}$.

However, if $\boldsymbol{h} \approx -\hat{\boldsymbol{h}}_c$, then an eigen-axis $\hat{\boldsymbol{e}}$ that is orthogonal to $\hat{\boldsymbol{h}}_c$ must be determined. Let the camera frame be defined through $\mathcal{C} : \{\hat{\boldsymbol{c}}_1, \hat{\boldsymbol{c}}_2, \hat{\boldsymbol{c}}_3\}$. The eigen-axis is determined first by taking a cross product with $\hat{\boldsymbol{c}}_1$:

$$\hat{\boldsymbol{e}}_{180} = \frac{\hat{\boldsymbol{h}}_c \times \hat{\boldsymbol{c}}_1}{|\hat{\boldsymbol{h}}_c \times \hat{\boldsymbol{c}}_1|} \tag{9}$$

If $\hat{\boldsymbol{h}}_c \approx \hat{\boldsymbol{c}}_1$, then this $\hat{\boldsymbol{e}}$ vector will have a small norm. In this ill-determined case, the $\hat{\boldsymbol{e}}$ vector is determined using

$$\hat{\boldsymbol{e}}_{180} = \frac{\hat{\boldsymbol{h}}_c \times \hat{\boldsymbol{c}}_2}{|\hat{\boldsymbol{h}}_c \times \hat{\boldsymbol{c}}_2|} \tag{10}$$

As $\hat{\boldsymbol{h}}_c$ cannot be aligned with both $\hat{\boldsymbol{c}}_1$ and $\hat{\boldsymbol{c}}_2$, this algorithm determines a unique eigen axis $\hat{\boldsymbol{e}}_{180}$ for the case that the principal rotation angle is close to 180 degrees, or $\pi - \Phi < \epsilon$. This special case eigen axis is only computed once in the module reset routine, is stored in camera frame components in `eHat180_C`, and is mapped into body frame components with Eq. (2) like the regular eigen axis.

In this scenario the angular velocity tracking error is evaluated using the same method as outlined in section 1.2.1.

2 Module Functions

- **Compute the attitude tracking error**: Determines the shortest rotation to align $\boldsymbol{h}$ and $\hat{\boldsymbol{h}}_c$, and computes the corresponding three-dimensional attitude difference
- **Control spacecraft rotation**: The reference frame is assumed to be non-accelerating and either zero (default) or a constant spin about the planet heading axis.
- **Robust to no planet heading information**: If the planet heading vector is not available, then the attitude feedback is turned off by zeroing $\boldsymbol{\sigma}_{B/R}$. Instead of driving the body rates to zero, the body rates are driven to a prescribed $\boldsymbol{\omega}_{R/N}$ vector. This allow the spacecraft to search for the planet and covers the case if some planet sensors are offline.
- **Robust to collinear observation vector**: The module must handle the cases where the commanded body relative vector and the planet direction vectors are nearly collinear.

3 Module Assumptions and Limitations

The module input vector $\boldsymbol{h}$ can be a vector of any length except for a zero-length vector.

4 Test Description and Success Criteria

The mathematics in this module are straight forward and can be tested in a series of input and output evaluation tests.

4.1 Check 1

Here a check is performed where the planet heading vector measurement $\boldsymbol{h}$ has a non-zero length and is not aligned with $\hat{\boldsymbol{h}}_c$.

4.2 Check 2

The planet direction vector measurement is marked as not valid and no previous valid image is available, so no saved heading exists. In this case the attitude tracking $\boldsymbol{\sigma}_{B/R}$ should be set to zero. Further, $\boldsymbol{\omega}_{R/N}$ is set to the prescribed search rate `omega_RN_B` and the body rate errors are evaluated as $\boldsymbol{\omega}_{B/R} = \boldsymbol{\omega}_{B/N} - \boldsymbol{\omega}_{R/N}$.

4.3 Check 3

The planet direction vector $\boldsymbol{h}$ aligned with $\hat{\boldsymbol{h}}_c$. In this case the attitude tracking $\boldsymbol{\sigma}_{B/R}$ should be set to zero. Further, the body rate errors are simply the inertial body angular rates.

4.4 Check 4

The planet direction vector is not given. The spacecraft is given a search rate as a sole command.

4.5 Non-Zero Camera Mounting

The camera frame is rotated relative to the body frame by several non-zero $\boldsymbol{\sigma}_{C/B}$ values and the spacecraft is placed at several attitudes $\boldsymbol{\sigma}_{B/N}$. The reference attitude implied by the output, $[RN] = [BR]^T[BN]$, must point $\hat{\boldsymbol{h}}_c$ at the target. The check is repeated after the measurement is dropped, so that the stored inertial heading is used. In this case the spacecraft is rotated to a different attitude $\boldsymbol{\sigma}_{B/N}$ between the measurement and the dropped update, so the stored heading only points $\hat{\boldsymbol{h}}_c$ at the target if it is kept in inertial components and mapped back with the current $[CN] = [CB][BN]$. The boresight-to-target angle of each update is evaluated with the attitude in effect at that update. The test `test_nonZeroCameraMounting` requires the boresight-to-target angle to be below 10^{-10} rad.

4.6 Antiparallel Heading

The target is placed directly behind $\hat{\boldsymbol{h}}_c$ for zero and non-zero $\boldsymbol{\sigma}_{C/B}$, with $\hat{\boldsymbol{h}}_c$ set to both $\hat{\boldsymbol{c}}_3$ and $\hat{\boldsymbol{c}}_1$ to exercise both cross products of the 180 degree fallback axis. The test `test_antiparallelHeading` requires the boresight-to-target angle of the reference attitude to be below 10^{-7} rad.

4.7 Saved Heading Timeout

Valid images are provided at selected times and invalid images otherwise, with the first image at 0 s and at 3 s. The saved heading must produce the tracking error of the last image for less than `timeOut` seconds after it, after which the module must output the search rate. An image that arrives exactly `timeOut` seconds after the previous one must be used on that update. The test `test_savedHeadingTimeout` checks $\boldsymbol{\sigma}_{B/R}$ and $\boldsymbol{\omega}_{R/N}$ to 10^{-12}.

4.8 Default Timeout

The module is run with `timeOut` left at its default value of zero, while valid and invalid images alternate starting with a valid image at 0 s. In this case each valid image is used once, on the update where it arrives, and the saved heading is discarded on the next update without a valid image, so it is never reused. Every valid update must output the tracking error of the image with a zero $\boldsymbol{\omega}_{R/N}$, and every invalid update must output a zero $\boldsymbol{\sigma}_{B/R}$ with the search rate. The test `test_defaultTimeOut` checks $\boldsymbol{\sigma}_{B/R}$ and $\boldsymbol{\omega}_{R/N}$ to 10^{-12}.

5 Test Parameters

The unit test verify that the module output guidance message vectors match expected values.

The nominal module test input values are $\hat{\boldsymbol{h}}_c = (0, 0, 1)$, $\boldsymbol{h} = (1, 0, 0)$ and ${}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} = (0.01, 0.50, -0.20)$ rad/sec. The nominal body-fixed search rate is set to ${}^{\mathcal{B}}\boldsymbol{\omega}_{R/N} = (0.0, 0.0, 0.1)$ rad/sec. This rate is only used if no planet direction vector is available. The small angle is set to $\epsilon = 0.01$ degrees.

Table 2: Error tolerance for each test.

Output Value Tested	Tolerated Error
$\boldsymbol{\sigma}_{B/R}$	1e-12
$\boldsymbol{\omega}_{B/R}$	1e-12
$\boldsymbol{\omega}_{R/N}$	1e-12
$\dot{\boldsymbol{\omega}}_{R/N}$	1e-12

6 Test Results

All of the tests passed:

Table 3: Test results

Check	Pass/Fail
1	PASSED
2	PASSED
3	PASSED
4	PASSED

7 User Guide

7.1 Input/Output Messages

The module has 2 required input messages, and 1 output message:

- `attGuidanceOutMsg` – This output message, of type `AttGuidMsgPayload`, provide the attitude tracking errors and the reference frame states.
- `opnavDataInMsg` – This input message, of type `OpNavMsgPayload`, receives the planet heading vector $\boldsymbol{h}$
- `imuInMsg` – This input message, of type `NavAttMsgPayload`, receives the inertial angular body rates $\boldsymbol{\omega}_{B/N}$
- `cameraConfigInMsg` – This input message, of type `CameraConfigMsgPayload`, receives the camera specs, notably the DCM: $[CB]$

7.2 Module Parameters and States

The module has the following parameter that can be configured:

- `alignAxis_C` – [REQUIRED] This 3x1 array contains the commanded camera-relative vector $\hat{\boldsymbol{h}}_c$ that is to be aligned with the planet heading $\boldsymbol{h}$
- `minUnitMag` – This double is currently not used by the module and is only kept for compatibility. No norm threshold is applied to $\boldsymbol{h}$. The default `minUnitMag` value is zero.
- `omega_RN_B` – This vectors specifies the body-fixed search rate to rotate and search for the sun if no good target vector is visible. Default value is a zero vector.
- `smallAngle` – This double specifies what is considered close for $\boldsymbol{h}$ and $\hat{\boldsymbol{h}}_c$ to be collinear. Default value is zero.
- `timeOut` – Time in seconds for which the inertial heading of the last valid image is reused while no new valid image is available. The saved heading is discarded once this time is reached, and a new valid image is always used immediately. Default value is zero, which disables reuse of past images.

- opNavAxisSpinRate – Specifies the nominal spin rate, in rad/s, about the commanded camera axis alignAxis_C. This is only used if a heading solution is available. Default value is zero, which brings the spacecraft to rest.